

Laurine Marty

*PhD Student in Machine Learning and
Neurosciences*

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Research Profile

I am a PhD student working at the intersection of machine learning and neuroscience. I investigate how Deep-Learning and theoretical neuroscience can help uncover mechanistic connectivity and causal relationships from electrophysiological brain recordings.

More broadly, I am interested in how dynamical and geometric structures emerge from high-dimensional neural data. Ultimately, my goal is to bridge these insights with clinically relevant applications.

Education

- Sept 2025 – **PhD in Machine Learning and Neurosciences**, *TU Wien*, Vienna, Austria
Present Machine Learning Research Unit
Supervisor: Prof. Guillaume Bellec
Topic : Integrating insights from theoretical neuroscience to improve large-scale deep learning training on electrophysiological data
- 2023 – 2025 **Cogmaster – Master in Cognitive Sciences (Major in Modeling)**, *ENS – PSL*, Paris, France
2023 **Master in Computer Science**, *Indian Institute of Technology (IIT)*, Delhi, India
- 2020 – 2023 **Engineering Degree in Mathematics and Computational Engineering**, *IMT Atlantique*, Brest, France
- 2017 – 2020 **Preparatory Classes**, *Lycée Joffre*, Montpellier, France
Physics, Mathematics and Engineering Sciences

Research Experience

- 2024 – 2025 **Research Assistant & Internship in Computational Neuroscience**, *Institute of Neuromodulation – UNICOG*, Paris, France
Supervisor: Joao Barbosa
○ Quantification of inter-individual differences for neural representations
○ Processing and Alignment EEG and fMRI recordings recorded simultaneously
○ Worked with large pretrained foundation models and transformer architectures.
- 2024 **Research Internship in Computational Neuroscience**, *ISIR – Sorbonne Université*, Paris, France
Supervisor: Benoit Girard
○ Applied reinforcement learning rules such as Opponent Actor Learning (OpAL) to a basal ganglia computational model.
○ Trained biologically plausible reinforcement learning models of rhesus monkey basal ganglia.
- 2023 **Research Internship in Cognitive Neuroscience**, *Lyfe Institute Research Center*, Lyon, France
Supervisor: Anne-Lise Saive
- 2022 **Research Engineer Internship in Machine Learning**, *Thales SIX GTS France*, Gif-sur-Yvette, France
○ Investigated how artificial intelligence can help measure and develop human creativity in culinary experts.
○ Developed few-shot learning methods for video classification of human behaviors.
○ Built a model combining dynamic and visual information from video inputs.
○ Created an embedded platform to deploy the prototype using Docker and Flask.

Publications

2026 **Self-supervised EEG pretraining amplifies individual differences in neural representations.**
Marty L., Pereira M., Bellec G., Barbosa J. *ICLR 2026 (Workshop)*

Technical Skills

Programming Python (PyTorch) , Julia, Java)

Machine Learning Deep learning, Self-Supervised, Contrastive learning, Reinforcement learning, Graph Learning

Neuroscience EEG, fMRI, Ephys data, Theoretical neurosciences models, Large-Scale Neural Simulations

Mathematics Optimization, Probability theory, linear algebra, dynamical systems, Dimensionality Reduction

Teaching and Academic Service

Summer 2026 **Lecturer**, *Advanced Deep Learning*, TU Wien

Winter 2025 **Lecturer**, *Deep Learning for Neuroscience*, TU Wien

3 years **Private Tutor**, *Mathematics, Physics and Computer Science*

Languages

French Native

English Fluent

German "I'm trying"

Research Interests

Computational neuroscience, neural data analysis, deep learning for brain recordings, causal inference, mechanistic modeling of neural connectivity.

References

Available upon request.